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GREEN CREMATION

Abstract

Systems for cremating a body. The system includes a box assembly and a vessel. The box assembly is for storing the body or a portion of a body. The box assembly includes a box defining a box cavity, a bottom end, a top end, and at least one side wall extending from the bottom end to the top end. The box assembly also includes an air conditioner disposed in the box cavity. The air conditioner is configured to control a temperature level or a humidity level in the box cavity. The vessel is coupled to the at least one side wall of the box. The vessel includes insect larvae that self-harvest, exit the vessel, and enter the box assembly to decompose the body or the portion of the body stored in the box assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of and claims priority to and the benefit of U.S. patent application Ser. No. 18/299,872 filed on Apr. 13, 2023 and titled “Green Cremation” (Attorney Docket No. 99798-101), which claims priority to and the benefit of U.S. Provisional Application No. 63/330,580 filed Apr. 13, 2022, titled “GREEN CREMATION,” the entire disclosures of which are hereby incorporated by reference for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to green cremation and more specifically, green cremation using insect larvae.

BACKGROUND

[0003] Over the years, flame-based cremations have surpassed burials as a primary choice to dispose deceased bodies, including human and animal bodies. Although this type of cremation is less harmful than burials, cremation still has negative impacts on the environment. For example, because flame-based cremation typically requires fuel to cremate bodies, it releases a considerable amount of carbon dioxide into the air. Therefore, there is a need for a more gentle, eco-friendly, and carbon-neutral method for cremation.

SUMMARY

[0004] Disclosed herein are implementations of a green cremation system. The present disclosure provides a system for cremating a body including, in one implementation, a box assembly and a vessel. The system may be situated in an environment whose conditions are controllable. The box assembly is for storing the body or a portion of the body. The box assembly includes a box and an air conditioner. The box defines a box cavity within the box, a bottom end, a top end defining a box opening connected to the box cavity, and at least one side wall extending from the bottom end to the top end. The air conditioner is disposed within the box cavity. The air conditioner is configured to control a temperature level or a humidity level in the box cavity when the air conditioner is in operation. The vessel is coupled to the at least one side wall of the box of the box assembly. The vessel includes insect larvae that self-harvest, exit the vessel, and enter the box assembly to decompose the body or the portion of the body stored in the box assembly.

[0005] The present disclosure also provides a system for cremating a body including, in one implementation, a box assembly, a vessel, and a cover. The box assembly is for storing the body or a portion of the body. The box assembly includes a box and an air conditioner. The box defines a box cavity within the box, a bottom end, a top end defining a box opening connected to the box cavity, and at least one side wall extending from the bottom end to the top end. The air conditioner is disposed within the box cavity. The air conditioner is configured to control a temperature level or a humidity level in the box cavity when the air conditioner is in operation. The vessel is coupled to the at least one side wall of the box of the box assembly. The vessel includes insect larvae that self-harvest, exit the vessel, and enter the box assembly to decompose the body or the portion of the body stored in the box assembly. The cover is hingedly connected to or removably disposed onto the top end of the box of the box assembly. The system is in an open state when the cover uncovers the box cavity of the box. The system is a closed state when the cover covers the box cavity of the box.

[0006] Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] For a more complete understanding of the present disclosure and its advantages, reference is now made to the following description, taken in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not necessarily to-scale. On the contrary, the dimensions of the various features may be—and typically are—arbitrarily expanded or reduced for the purpose of clarity.

[0008] FIG. 1 depicts a schematic perspective view of a system for green cremation according to a first embodiment of the present disclosure.

[0009] FIG. 2 depicts an embodiment of a composting bin disposed in the vessel as described in FIG. 1.

[0010] FIG. 3A depicts a schematic perspective view of the box assembly of the system as described in FIG. 1.

[0011] FIG. 3B depicts a schematic perspective view of the vessel as described in FIG. 1.

[0012] FIG. 3C depicts a schematic isometric view of the box assembly of the system as described in FIG. 1.

[0013] FIG. 4 depicts a schematic perspective view of a system for green cremation according to a second embodiment of the present disclosure.

[0014] FIG. 5 depicts a schematic perspective view of a system for green cremation according to a third embodiment of the present disclosure.

[0015] FIG. 6A depicts a schematic perspective view of the box assembly of the system as described in FIG. 5.

[0016] FIG. 6B depicts a schematic perspective view of the vessel as described in FIG. 5.

[0017] FIG. 6C depicts a schematic isometric view of the box assembly of the system as described in FIG. 5.

[0018] FIG. 7 depicts a schematic perspective view of a system for green cremation according to a fourth embodiment of the present disclosure.

NOTATION AND NOMENCLATURE

[0019] Various terms are used to refer to particular system components. A particular component may be referred to commercially or otherwise by different names. Further, a particular component (or the same or similar component) may be referred to commercially or otherwise by different names. Consistent with this, nothing in the present disclosure shall be deemed to distinguish between components that differ only in name but not in function. In the following discussion and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . .” Also, the term “couple” or “couples” is intended to mean either an indirect or direct connection. Thus, if a first device couples to a second device, that connection may be through a direct connection, or through an indirect connection via other devices and connections.

[0020] The terminology used herein is for the purpose of describing particular example implementations only, and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

[0021] The terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections; however, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish

one element, component, region, layer, or section from another region, layer, or section. Terms such as “first,” “second,” and other numerical terms, when used herein, do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of the example implementations. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C. In another example, the phrase “one or more” when used with a list of items means there may be one item or any suitable number of items exceeding one.

[0022] Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,” “up,” “upper,” “top,” “bottom,” “down,” “inside,” “outside,” “contained within,” “superimposing upon,” and the like, may be used herein. These spatially relative terms can be used for ease of description to describe one element's or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms may also be intended to encompass different orientations of the device in use, or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptions used herein interpreted accordingly.

DETAILED DESCRIPTION

[0023] The following discussion is directed to various implementations of the present disclosure. Although one or more of these implementations may be preferred, the implementations disclosed should not be interpreted, or otherwise used, as limiting the scope of the present disclosure, including the claims. In addition, one skilled in the art will understand that the following description has broad application, and the discussion of any implementation is meant only to be exemplary of that implementation, and not intended to intimate that the scope of the disclosure, including the claims, is limited to that implementation.

[0024] Aspects of the present disclosure relates to green cremation and more specifically, green cremation using insect larvae. In some implementations, the insect larvae may be black soldier fly larvae (BSFL). FIG. 1 depicts a schematic perspective view of a system for green cremation according to a first embodiment of the present disclosure. The system **10** may be situated in an environment whose conditions are controllable. The conditions of the environment may include parameters such as a temperature level, a humidity level, a moisture level, odors, and an amount of air. For example, the environment may be controlled to maintain at a certain temperature and/or humidity level as desired. In FIG. 1, the system **10** may include a box assembly **12** for storing at least one body or a portion of a body. The body may be a human body, a non-human (e.g., dog or cat) body, or an organ from a human or non-human body. The box assembly **12** may include a box **14** defining a box cavity **15** within the box **14**. The box **14** may further include a bottom end **13** (e.g., a base), a top end **17** defining a box opening connected to the box cavity **15**, and at least one side wall (e.g., side wall **16**) extending from the bottom end **13** to the top end **17**. The side wall **16** at least partially defines the box cavity **15**. The side wall **16** may be made of a thermal insulation material. Although FIG. 1 depicts that the shape of the box assembly **12** is rectangular, it is contemplated that in one or more implementations of the present disclosure, the box assembly **12** may be in any other desired geometric shapes, including, but not limited to, round, square, oval, or polygonal. The box assembly **12** may further include a size to receive a body or a portion of the body therein.

[0025] As shown in FIG. 1, the box assembly **12** may further include an insert **18** disposed within the box cavity **15**, which separates the box cavity **15** into a first chamber **19** and a second chamber

20. The insert **18** may be disposed at any location within the box cavity **15**. In some implementations, the insert **18** may be connected to the side wall **16** of the box **14**. In some other implementations, the insert **18** may be connectable to the side wall **16** of the box **14**, in which case the insert **18** may be removable or replaceable. As shown in FIG. **1**, the insert **18** may include an airway **21** through which air in the box cavity **15** may flow from the first chamber **19** to the second chamber **20**, or vice versa. The airway **21** may be an opening disposed in the insert **18**. In some implementations, a vent may be used to cover the opening and allow for the air in the box cavity **15** to flow between the first chamber **19** and the second chamber **20**. The box assembly **12** may further include an air conditioner **32** (see FIG. **3C**) disposed in the first chamber **19**. The air conditioner **32** is configured to control a temperature level and/or a humidity level in the box cavity **15** when it is in operation. In some implementations, the air conditioner **32** may include one or more devices that control other environmental conditions within the box cavity **15** (e.g., moisture control and odors). The second chamber **20** may include a shape and a size to receive a body or a portion of the body therein for cremation as described herein.

[0026] Referring to FIG. **1**, the box assembly **12** may further include an air vent **22** disposed on the side wall **16** of the box **14** thereof. The air vent **22** may be configured for air to be exchanged between the box cavity **15** and the environment in which the system is situated. For example, the air vent **22** may be configured for the air inside the box cavity **15** to exit the box assembly **12** or may be configured for the air in the environment to enter the box assembly **12**.

[0027] In some implementations, the box assembly **12** may further include an ozone generator disposed on the side wall **16** of the box **14**. The ozone generator (not shown) may be configured to remove odors from the box cavity **15**. For example, the ozone generator may be disposed in an exhaust chamber with a one-way air valve. In such an implementation, the ozone generator is configured to remove odors from the box assembly **12** without exposing the insects in the box assembly **12** to ozone.

[0028] The box assembly **12** may also include an electronic control panel **24** connected thereto and configured to control the operation of the system **10**, including the operation of the air conditioner **32**. In some implementations, the electronic control panel **24** may be connected to the side wall **16** of the box **14**. According to one or more implementations of the present disclosure, the electronic control panel **24** may be a display interface, through which an operator may input operating instructions to the system **10** for the purpose of controlling the conditions (e.g., a temperature level, a humidity level, or an air amount) within the box assembly **12**. Upon a request by the operator, the electronic control panel **24** may display images or videos showing a scene of the box cavity **15**. This allows the operator to monitor the operation of the system **10** in real time or near real time. The electronic control panel **24** may further electronically communicate with a remote server, such as a remote storage cloud, a remote control center, or a mobile phone of the operator or of any person of interest (e.g., a family member of the at least one body or portion(s) of the body). The electronic control panel **24** may also, by wires or wirelessly, connect to at least one another electronic control panel of another system of this kind. The connection between or among such systems may provide additional features, including, but not limited to, data sharing between or among the systems, remote controls of the systems, or alert notifications among the systems. For example, if there is a technical problem with one of the connected systems, the system having the technical problem may generate an alert, which may be sent to other connected systems or to a technician or an operator who may then fix the technical problem.

[0029] The electronic control panel **24** may also leverage the Internet of Things (IoT), Bluetooth, or Wi-Fi to remotely connect to, enable, and control one or more functionalities of the system **10**. For example, the electronic control panel **24** may be configured to present a display interface on a remote device, through which a user may input operating instructions to the system **10** for the purpose of controlling the conditions within the box assembly **12** (e.g., a temperature, a humidity level, or an air amount).

[0030] To carry out green cremation according to this implementation of the present disclosure, the system **10** may further include a vessel **26** coupled to the side wall **16** of the box assembly **12**. In some implementations, the vessel **26** may be glued, screwed, or hooked to the side wall **16** of the box assembly **12**, or any other desired coupling mechanisms.

[0031] As shown in FIG. **1**, the vessel **26** may be coupled to an outer surface of the side wall **16** of the box assembly **12**. In some implementations, the vessel **26** may be integrated with the box assembly **12**. In some other implementations, the vessel **26** may be removably attached to the box assembly **12**. It is contemplated that although FIG. **1** shows one vessel coupled to the side wall **16** of the box assembly **12**, more than one vessels may be similarly coupled thereto. The box assembly **12** may include a first orifice **29** (see FIG. **3A**) on the side wall **16** thereof. The vessel **26** may include a second orifice **30** (see FIG. **3B**), which may, with the first orifice **29**, form a channel between the vessel **26** and the box assembly **12** when the vessel **26** is coupled to the box assembly **12**. It is further contemplated that when more than one vessels coupled to the side wall **16** of the box assembly **12**, the box assembly **12** may include more than one orifices on the side wall **16** thereof to accommodate the orifices on those vessels. In some other implementations, the vessel **26** may include more than one orifices in addition to the second orifice **30**.

[0032] The vessel **26** may contain BSFL. The BSFL are hatched from eggs laid by adult female black soldier flies (BSF). After hatching, a larval development stage may begin, during which the BSFL may consume organic or inorganic matters (e.g., organic or inorganic waste in the vessel **26**). The larval development stage may last about two or three weeks. Thereafter, the BSFL may mature into juvenile and/or adult larvae, which may later turn into adult black soldier flies (BSF). The adult BSF may then mate (e.g., in the air within the vessel **26**), and adult female BSF may lay more eggs to harvest another generation of BSFL. The adult BSF may live for a few more days, or a week or more, until they die.

[0033] The vessel **26** may contain, in addition to or independent from the BSFL, insect larvae from other insects including but not limited to blowflies, flesh flies, scorpion flies, dermestid beetles, carrion beetles, rove beetles, hister beetles, noctuid caterpillar, or a combination thereof. After hatching, a larval development stage may begin, during which the insect larvae may consume organic or inorganic matters (e.g., organic or inorganic waste in the vessel **26**). Thereafter, the insect larvae may mature into juvenile and/or adult larvae, which may later turn into adult insects. The adult insects may then mate (e.g., in the air within the vessel **26**), and adult female insects may lay more eggs to harvest another generation of insect larvae. For ease of understanding, the description describes green cremation using BSFL. However, it is to be understood that insect larvae from other insects, such as the ones described above, may be used.

[0034] The vessel **26** may include a shape and a size to allow the BSFL to self-harvest there within, as described herein. For example, the vessel **26** may include a shape and a size to allow the BSFL to establish one or more colonies there within. The vessel **26** may include a slanted wall having a slope angle so as to allow the BSFL to exit the vessel **26**. If the slope angle is steep, the BSFL may not easily exit the vessel **26**. On the other hand, if the slope angle is shallow, the BSFL may exit the vessel **26** at an early stage during the larval development stage, which is not helpful for establishing the colonies within the vessel **26**. In some implementations, the slope angle may be between approximately 35° and 45°. Referring to FIG. **1**, when the vessel **26** is coupled to the box assembly **12**, the BSFL may exit the vessel **26** and enter the box assembly **12**. If there is a body or a portion of a body in the box assembly **12**, the BSFL may begin to decompose the body or the portion of the body. Referring to FIG. **1**, when the vessel **26** is coupled to the box assembly **12**, the BSFL may exit the vessel **26** and enter the box assembly **12**. If there is a body or a portion of a body in the box assembly **12**, the BSFL may begin to decompose the body or the portion of the body. To escape the heat produced by the decomposing body, the BSFL may be motivated to exit the second chamber **20** of the box cavity **15**. In some implementations, the second chamber **20** has slanted walls with a slope angle so as to allow the BSFL to escape the second chamber **20** and then be collected in the

bottom of the first chamber **19** of the box cavity **15**

[0035] In some implementations, the vessel **26** may include multiple structures. For example, the BSFL may self-harvest from a first structure of the vessel **26** into a second vessel of the vessel **26** using an escape ramp having an angle of no more than 60°. The BSFL may then exit the second structure and enter the box assembly **12**.

[0036] In some implementations, the vessel **26** may include a composting bin disposed therein for harvesting the BSFL. FIG. 2 depicts an example of the composting bin disposed in the vessel **26**. The composting bin **28** may include dual walls for retaining heat in the vessel **26**. An example of a design for the composting bin **28** is described in U.S. application Ser. No. 29/624,567, the disclosure of which is hereby incorporated by reference in its entirety. The composting bin **28** and/or the vessel **26** may be in other shapes or sizes. In some implementations, the composting bin may be the vessel **26** as described herein.

[0037] The vessel **26** may have a controlled temperature level and/or humidity level suitable for the BSFL to self-harvest there within. In some implementations, the temperature level in the vessel **26** may be controlled to be between approximately 75 degrees Fahrenheit (° F.) and 95° F., or any other desired temperature levels. In other implementations, the temperature level in the vessel **26** may be controlled to be between approximately 65° F. and 100° F.

[0038] Referring to FIG. 1, when the vessel **26** is coupled to the box assembly **12**, an operator may adjust the conditions (e.g., a temperature level, a humidity level, and/or an amount of air) inside the box assembly **12** as well as in the vessel **26** via the electronic control panel **24** so as to provide a suitable condition for the BSFL to self-harvest in the vessel **26**. In some implementations, the temperature level may be set in a range of approximately 65° F. and 100° F., or at any other desired temperature levels. The BSFL may, out of its nature, self-harvest in the vessel **26**. The BSFL may exit the vessel **26** and enter the box assembly **12** through the channel created by the first orifice **29** and the second orifice **30**. When the BSFL are present in the box assembly **12**, they may begin to decompose the at least one body or a portion of the body stored in the box assembly **12**. The BSFL may consume bone marrows of the body or the portion of the body. Depending on the amount of the BSFL available in the box assembly **12**, the conditions within the box assembly **12**, and the size of the body or the portion of the body, the decomposition may last for a few days. After the decomposition is completed, remains of the body or the portion of the body may be removed from the box assembly **12** and returned to family members. The remains may also be donated to science for research studies, planted in the ground (e.g., in a garden), or fed to chicken, fish, birds, or the like. Comparing with the traditional cremation that uses fire, the system **10** described herein for decomposing bodies offers green cremation, which is a gentle, eco-friendly, and carbon-neutral alternative to the flame-based cremation.

[0039] Although the system **10** as described herein includes one box assembly for green cremation as illustrated herein, more than one such box assemblies may be connected, either side-by-side or end-to-end, when in use. When more than one such box assemblies are connected, each box assembly may use its own electronic control panel and air conditioner to control the conditions there within as well as the conditions in any vessels coupled thereto. Alternatively, the conditions of several such connected box assemblies and any vessels coupled thereto may be controlled by a common electronic control panel and a common air conditioner. Furthermore, when more than one such box assemblies are connected, each box assembly may be coupled by at least one vessel containing BSFL. Alternatively, several such connected box assemblies may share the BSFL coming from one or more common vessel coupled to those box assemblies.

[0040] FIG. 3A depicts a schematic perspective view of the box assembly of the system as described in FIG. 1. As shown in FIG. 3A, the first orifice **29** is provided on the side wall **16** of the box assembly **12**. The first orifice **29** may be configured for the BSFL to enter the box assembly **12** for decomposing a body or a portion of a body. As discussed in FIG. 1, when more than one vessels are coupled to the side wall **16** of the box assembly **12**, the box assembly **12** may include more than

one orifices on the side wall **16** thereof to accommodate the orifices on those vessels.

[0041] FIG. **3B** depicts a schematic perspective view of the vessel as described in FIG. **1**. The vessel **26** may contain BSFL. The vessel **26** may include a shape and a size to allow the BSFL to self-harvest there within. For example, the vessel **26** may include a shape and a size to allow the BSFL to establish one or more colonies there within. The vessel **26** may include a slanted wall having a slope angle so as to allow the BSFL to exit the vessel **26**. In some implementations, the slope angle may be between approximately 35° and 45°. When the vessel **26** is coupled to the box assembly **12**, the BSFL may exit the vessel **26** and enter the box assembly **12**. If there is a body or a portion of a body in the box assembly **12**, the BSFL may begin to decompose the body or the portion of the body. In one or more implementations, the vessel **26** may include a composting bin disposed therein for harvesting the BSFL, as illustrated in FIG. **2**. In some implementations, the composting bin may be the vessel **26** as described herein.

[0042] FIG. **3B** shows that the second orifice **30** is provided on the vessel **26**. When the vessel **26** is coupled to the box assembly **12**, the BSFL may exit the vessel **26** via the second orifice **30** and enter the box assembly **12** through the first orifice **29** provided on the at least side wall **16** of the box assembly **12**. As discussed in FIG. **1**, the vessel **26** may include more than one orifices in addition to the second orifice **30**.

[0043] FIG. **3C** depicts a schematic isometric view of the box assembly of the system as described in FIG. **1**. As shown in FIG. **3C**, the insert **18** may be disposed within the box cavity **15**, which separates the box cavity **15** into the first chamber **19** and the second chamber **20**. The insert **18** may be disposed at any location within the box cavity **15**. In some implementations, the insert **18** may be connected to the side wall **16** of the box **14**. In some other implementations, the insert **18** may be connectable to the side wall **16** of the box **14**, in which case the insert **18** may be removable or replaceable. FIG. **3C** further shows that the air conditioner **32** may be disposed in the first chamber **19**. The air conditioner **32** may be configured to control a temperature level and/or humidity level in the box cavity **15** when the air conditioner **32** is in operation. For example, during the decomposition process, the temperature level within the box **14** may rise. The air conditioner **32** may thus be controlled to operate to maintain the temperature level within the desired temperature range as described herein. The second chamber **20** may include a shape and a size to receive a body or a portion of a body for green cremation. In addition, FIG. **3C** shows that the first orifice **29** is provided on the side wall **16** of the box assembly **12** for the BSFL entering the box assembly **12** from the vessel **26** when the vessel **26** is coupled to the side wall **16** of the box assembly **12**.

[0044] FIG. **4** depicts a schematic perspective view of a system for green cremation according to a second implementation of the present disclosure. The system **50** may be positioned in an environment whose conditions are controllable. The conditions of the environment may include parameters such as a temperature level, a humidity level, a moisture level, and odors. For example, the environment may be controlled to maintain at a certain temperature and/or humidity level as desired. In FIG. **4**, the system **50** may include a box assembly **52** for storing a body or a portion of a body. The body may be a human body, a non-human (e.g., dog or cat) body, or an organ from a human or non-human body. The box assembly **52** may include a box **54** defining a box cavity **55** within the box. The box **54** may further include a bottom end **53** (e.g., a base), a top end **57** defining a box opening connected to the box cavity **55**, and at least one side wall (e.g., side wall **56**) extending from the bottom end **53** to the top end **57**. The side wall **56** at least partially defines the box cavity **55**. The side wall **56** may be made of a thermal insulation material. Although FIG. **4** depicts that the shape of the box assembly **52** is rectangular, it is contemplated that in one or more implementations of the present disclosure, the box assembly **52** may be in any other desired geometric shapes, including, but not limited to, round, square, oval, or polygonal. The box assembly **52** may further include a size to receive a body or a portion of the body therein.

[0045] As shown in FIG. **4**, the box assembly **52** may further include an insert **58** disposed within the box cavity **55**, which separates the box cavity **55** into a first chamber **59** and a second chamber

60. The insert **58** may be disposed at any location within the box cavity **55**. In some implementations, the insert **58** may be connected to the side wall **56** of the box **54**. In some other implementations, the insert may be connectable to the side wall **56** of the box **54**, in which case the insert **58** may be removable or replaceable. As shown in FIG. **4**, the insert **58** may include an airway **61** through which air in the box cavity **55** may flow from the first chamber **59** to the second chamber **60**, or vice versa. The airway **61** may be an opening disposed in the insert **58**. In some implementations, a vent may be used to cover the opening and allow for the air in the box cavity **55** to flow between the first chamber **59** and the second chamber **60**. The box assembly **52** may further include an air conditioner (similar to **32** in FIG. **3C**) disposed in the first chamber **59**. The air conditioner may be configured to control a temperature level and/or a humidity level in the box cavity **55** when it is in operation. In some implementations, the air conditioner may include one or more devices that control other environmental conditions within the box cavity **55** (e.g., moisture control and odors). The second chamber **60** may include a shape and a size to receive a body or a portion of a body for green cremation.

[0046] Referring to FIG. **4**, the box assembly **52** may further include an air vent **62** disposed on the side wall **56** of the box **54** thereof. The air vent **62** may allow air to be exchanged between the box cavity **55** and the environment in which the system is situated. For example, the air vent **62** may be configured for the air inside the box cavity **55** to exit the box assembly **52** or may be configured for the air in the environment to enter the box assembly **52**.

[0047] In some implementations, the box assembly **52** may further include an ozone generator disposed on the side wall **56** of the box **54**. The ozone generator (not shown) may be configured to remove odors from the box cavity **55**. For example, the ozone generator may be disposed in an exhaust chamber with a one-way air valve. In such an implementation, the ozone generator is configured to remove odors from the box assembly **52** without exposing the insects in the box assembly **52** to ozone.

[0048] The box assembly **52** may also include an electronic control panel **64** connected thereto and configured to control the operation of the system **50**, including the operation of the air conditioner. In some implementations, the electronic control panel **64** may be connected to the side wall **56** of the box **54**. According to one or more implementations of the present disclosure, the electronic control panel **64** may be a display interface, through which an operator may input operating instructions to the system **50** for the purpose of controlling the conditions (e.g., a temperature level, a humidity level, or an air amount) within the box assembly **52**. Upon a request by the operator, the electronic control panel **64** may display images or videos showing a scene of the box cavity **55**. This allows the operator to monitor the operation of the system **50** in real time or near real time. The electronic control panel **64** may further electronically communicate with a remote server, such as a remote storage cloud, a remote control center, or a mobile phone of the operator or of any person of interest (e.g., a family member of the body or the portion of the body). The electronic control panel **64** may also, by wires or wirelessly, connect to at least one another electronic control panel of another system of this kind. The connection between or among such systems may provide additional features, including, but not limited to, data sharing between or among the systems, remote controls of the systems, or alert notifications among the systems. For example, if there is a technical problem with one of the connected systems, the system having the technical problem may generate an alert, which may be sent to other connected systems or to a technician or an operator who may then fix the technical problem.

[0049] The electronic control panel **64** may also leverage the Internet of Things (IoT), Bluetooth, or Wi-Fi to remotely connect to, enable, and control one or more functionalities of the system **50**. For example, the electronic control panel **64** may be configured to present a display interface on a remote device, through which a user may input operating instructions to the system **50** for the purpose of controlling the conditions within the box assembly **52** (e.g., a temperature, a humidity level, or an air amount).

[0050] To carry out green cremation according to this embodiment of the present disclosure, the system **50** may further include a vessel **66** coupled to the side wall **56** of the box assembly **52**. In some implementations, the vessel **66** may be glued, screwed, or hooked to the side wall **56** of the box assembly **52**, or any other desired coupling mechanisms.

[0051] As shown in FIG. **4**, the vessel **66** may be coupled to an inner surface of the side wall **56** of the box assembly **52**. In some implementations, the vessel **66** may be integrated with the box assembly **52**. In some other implementations, the vessel **66** may be removably attached to the box assembly **52**. It is contemplated that although FIG. **4** shows one vessel coupled to the side wall **56** of the box assembly **52**, more than one vessels may be similarly coupled thereto. It is further contemplated that although FIG. **4** shows that the vessel **66** is coupled into the second chamber **60**, the vessel **66** may also be coupled into the first chamber **59**. The vessel **66** may include an orifice **68**, which may be configured for BSFL to exit the vessel **66** to decompose the body or the portion of the body stored in the box assembly **52**. In some implementations, the vessel **66** may include more than one orifices in addition to the orifice **68** shown in FIG. **4**.

[0052] The vessel **66** may contain BSFL. The vessel **66** may include a shape and a size to allow the BSFL to self-harvest there within, as described herein. For example, the vessel **66** may include a shape and a size to allow the BSFL to establish one or more colonies there within. The vessel **66** may include a slanted wall having a slope angle so as to allow the BSFL to exit the vessel **66**. In some implementations, the slope angle may be between approximately 35° and 45°. Referring to FIG. **4**, when the vessel **66** is coupled to the box assembly **52**, the BSFL may exit the vessel **66** and enter the box assembly **52**. If there is a body or a portion of a body in the box assembly **52**, the BSFL may begin to decompose the body or the portion of the body. To escape the heat produced by the decomposing body, the BSFL may be motivated to exit the second chamber **60** of the box cavity **55**. In some implementations, the second chamber **60** has slanted walls with a slope angle so as to allow the BSFL to escape the second chamber **60** and then be collected in the bottom of the first chamber **59** of the box cavity **55**.

[0053] In some implementations, the vessel **66** may include multiple structures. For example, the BSFL may self-harvest from a first structure of the vessel **66** into a second vessel of the vessel **66** using an escape ramp having an angle of no more than 60°. The BSFL may then exit the second structure and enter the box assembly **52**.

[0054] In some implementations, the vessel **66** may include a composting bin disposed therein for harvesting the BSFL. The composting bin may include dual walls for retaining heat in the vessel **26**, as illustrated in FIG. **2**. The composting bin and/or the vessel **66** may be in other shapes or sizes. In some implementations, the composting bin may be the vessel **66** as described herein.

[0055] The vessel **66** may have a controlled temperature level and/or humidity level suitable for the BSFL to self-harvest there within. In some implementations, the temperature level in the vessel **66** may be controlled to be between approximately 75° F. and 95° F., or any other desired temperature levels. In other implementations, the temperature level in the vessel **66** may be controlled to be between approximately 65° F. and 100° F.

[0056] Referring to FIG. **4**, when the vessel **66** is coupled to the box assembly **52**, an operator may adjust the conditions (e.g., a temperature level, a humidity level, and/or an amount of air) inside the box assembly **52** as well as in the vessel **66** via the electronic control panel **64** so as to provide a suitable condition for the BSFL to self-harvest in the vessel **66**. In some implementations, the temperature level may be set in a range of approximately 65° F. and 100° F., or at any other desired temperature levels. The BSFL may, out of its nature, self-harvest in the vessel **66**. The BSFL may exit the vessel **66** and enter the box assembly **52** through the orifice **68**. When the BSFL are present in the box assembly **52**, they may begin to decompose the body or the portion of the body stored in the box assembly **52**. The BSFL may consume bone marrows of the body or the portion of the body. Depending on the amount of the BSFL available in the box assembly **52**, the conditions within the box assembly **52**, and the size of the body or the portion of the body, the decomposition

may last for a few days or a week or more. After the decomposition is completed, remains of the body or the portion of the body may be removed from the box assembly **52** and returned to family members. The remains may also be donated to science for research studies, planted in the ground (e.g., in a garden), or fed to chicken, fish, birds, or the like. Comparing with the traditional cremation that uses fire, the system **50** described herein for decomposing bodies offers green cremation, which is a gentle, eco-friendly, and carbon-neutral alternative to the flame-based cremation.

[0057] Although the system **50** as described herein includes one box assembly for green cremation as illustrated herein, more than one such box assemblies may be connected, either side-by-side or end-to-end, when in use. When more than one such box assemblies are connected, each box assembly may use its own electronic control panel and air conditioner to control the conditions there within as well as the conditions in any vessels coupled thereto. Alternatively, the conditions of several such connected box assemblies and any vessels coupled thereto may be controlled by a common electronic control panel and a common air conditioner. Furthermore, when more than one such box assemblies are connected, each box assembly may be coupled to at least one vessel containing BSFL. Alternatively, several such connected box assemblies may share the BSFL coming from one or more common vessel coupled to those box assemblies.

[0058] FIG. 5 depicts a schematic perspective view of a system for green cremation according to a third implementation of the present disclosure. As shown in FIG. 5, the system **80** may include a box assembly **82** for storing a body or a portion of a body. The body may be a human body, a non-human (e.g., dog or cat) body, or an organ from a human or non-human body. The box assembly **82** may include a box **84** defining a box cavity **85** within the box. The box **84** may further include a bottom end **83** (e.g., a base), a top end **87** defining a box opening connected to the box cavity **85**, and at least one side wall (e.g., side wall **86**) extending from the bottom end **83** to the top end **87**. The side wall **86** at least partially defines the box cavity **85**. The side wall **86** may be made of a thermal insulation material. Although FIG. 5 depicts that the shape of the box assembly **82** is rectangular, it is contemplated that in one or more implementations of the present disclosure, the box assembly **82** may be in any other desired geometric shapes, including, but not limited to, round, square, oval, or polygonal. The box assembly **82** may further include a size to receive a body or a portion of a body therein.

[0059] As shown in FIG. 5, the box assembly **82** may further include an insert **88** disposed within the box cavity **85**, which separates the box cavity **85** into a first chamber **89** and a second chamber **90**. The insert **88** may be disposed at any location within the box cavity **85**. In some implementations, the insert **88** may be connected to the side wall **86** of the box **84**. In some other implementations, the insert **88** may be connectable to the side wall **86** of the box **84**, in which case the insert **88** may be removable or replaceable. As shown in FIG. 5, the insert **88** may include an airway **91** through which air in the box cavity **85** may flow from the first chamber **89** to the second chamber **90**, or vice versa. The airway **91** may be an opening disposed in the insert **88**. In some implementations, a vent may be used to cover the opening and allow for the air in the box cavity **85** to flow between the first chamber **89** and the second chamber **90**. The box assembly **82** may further include an air conditioner **104** (see FIG. 6C) disposed in the first chamber **89**. The air conditioner **104** may be configured to control a temperature level and/or a humidity level in the box cavity **85** when it is in operation. In some implementations, the air conditioner **104** may include one or more devices that control other environmental conditions within the box cavity **85** (e.g., moisture control and odors). The second chamber **90** may include a shape and a size to receive a body or a portion of a body for green cremation.

[0060] Referring to FIG. 5, the box assembly **82** may further include an air vent **92** disposed on the side wall **86** of the box **84** thereof. The air vent **92** may allow air to be exchanged between the box cavity **85** and an ambient environment (i.e. outside the system **80**). For example, the air vent may be configured for the air inside the box cavity **85** to exit the box assembly **82** or may be configured

for the air in the ambient environment to enter the box assembly **82**.

[0061] In some implementations, the box assembly **82** may further include an ozone generator disposed on the side wall **86** of the box **84**. The ozone generator (not shown) may be configured to remove odor from the box cavity **85**. For example, the ozone generator may be disposed in an exhaust chamber with a one-way air valve. In such an implementation, the ozone generator is configured to remove odors from the box assembly **82** without exposing the insects in the box assembly **82** to ozone.

[0062] The box assembly **82** may also include an electronic control panel **94** connected thereto and configured to control the operation of the system **80**, including the operation of the air conditioner. In some implementations, the electronic control panel **94** may be connected to the side wall **86** of the box **84**. According to one or more implementations of the present disclosure, the electronic control panel **94** may be a display interface, through which an operator may input operating instructions to the system **80** for the purpose of controlling the conditions (e.g., a temperature level, a humidity level, or an air amount) within the box assembly **82**. Upon a request by the operator, the electronic control panel **94** may display images or videos showing a scene of the box cavity **85**. This allows the operator to monitor the operation of the system **80** in real time or near real time. The electronic control panel **94** may further electronically communicate with a remote server, such as a remote storage cloud, a remote control center, or a mobile phone of the operator or of any person of interest (e.g., a family member of the body or the portion of the body). The electronic control panel **94** may also, by wires or wirelessly, connect to at least one another electronic control panel of another system of this kind. The connection between or among such systems may provide additional features, including, but not limited to, data sharing between or among the systems, remote controls of the systems, or alert notifications among the systems. For example, if there is a technical problem with one of the connected systems, the system having the technical problem may generate an alert, which may be sent to other connected systems or to a technician or an operator who may then fix the technical problem.

[0063] The electronic control panel **94** may also leverage the Internet of Things (IoT), Bluetooth, or Wi-Fi to remotely connect to, enable, and control one or more functionalities of the system **80**. For example, the electronic control panel **94** may be configured to present a display interface on a remote device, through which a user may input operating instructions to the system **80** for the purpose of controlling the conditions within the box assembly **82** (e.g., a temperature, a humidity level, or an air amount).

[0064] Referring to FIG. 5, the system **80** may further include a cover **96** which may cover or uncover the box opening of the box **84** of the box assembly **82** thereof. In some implementations, the cover **96** may be hingedly connected to the top end **87** of the box **84**. In some other implementations, the cover **96** may be removably disposed onto the top end **87** of the box **84**. The system **80** may be in an open state when the cover **96** uncovers the box cavity **85**. When the system **80** is in the open state, the operator may place a body or a portion of a body into the box assembly **82** (e.g., in the second chamber). The system **80** may be in a closed state when the cover **96** covers the box cavity **85**. When the system **80** is in the closed state, the operator may input operating instructions to the system **80** to control the conditions (e.g., a temperature level, a humidity level, or an air amount) within the box assembly **82** for green cremation.

[0065] To carry out green cremation according to this embodiment of the present disclosure, the system **80** may further include a vessel **98** coupled to the side wall **86** of the box assembly **82**. In some implementations, the vessel **98** may be glued, screwed, or hooked to the side wall **86** of the box assembly, or any other desired coupling mechanisms.

[0066] As shown in FIG. 5, the vessel **98** may be coupled to an outer surface of the side wall **86** of the box assembly **82**. In some implementations, the vessel **98** may be integrated with the box assembly **82**. In some other implementations, the vessel **98** may be removably attached to the box assembly **82**. It is contemplated that although FIG. 5 shows one vessel coupled to the side wall **86**

of the box assembly **82**, more than one vessels may be similarly coupled thereto. The box assembly **82** may include a first orifice **100** (see FIG. **6A**) on the side wall **86** of the box **84** thereof. The vessel **98** may include a second orifice **102** (see FIG. **6B**), which may, with the first orifice **100**, form a channel between the vessel **98** and the box assembly **82** when the vessel **98** is coupled to the box assembly **82**. It is further contemplated that when more than one vessels coupled to the side wall **86** of the box assembly **82**, the box assembly **82** may include more than one orifices on the side wall **86** thereof to accommodate the orifices on those vessels. In some other implementations, the vessel **98** may include more than one orifices in addition to the second orifice **102**.

[0067] The vessel **98** may contain BSFL. The vessel **98** may include a shape and a size to allow the BSFL to self-harvest there within, as described herein. For example, the vessel **98** may include a shape and a size to allow the BSFL to establish one or more colonies there within. The vessel **98** may include a slanted wall having a slope angle so as to allow the BSFL to exit the vessel **98**. In some embodiments, the slope angle may be between approximately 35° and 45°. Referring to FIG. **5**, when the vessel **98** is coupled to the box assembly **82**, the BSFL may exit the vessel **98** and enter the box assembly **82**. If there is a body or a portion of a body in the box assembly **82**, the BSFL may begin to decompose the body or the portion of the body. To escape the heat produced by the decomposing body, the BSFL may be motivated to exit the second chamber **90** of the box cavity **85**. In some implementations, the second chamber **90** has slanted walls with a slope angle so as to allow the BSFL to escape the second chamber **90** and then be collected in the bottom of the first chamber **89** of the box cavity **85**.

[0068] In some implementations, the vessel **98** may include multiple structures. For example, the BSFL may self-harvest from a first structure of the vessel **98** into a second vessel of the vessel **98** using an escape ramp having an angle of no more than 60°. The BSFL may then exit the second structure and enter the box assembly **82**.

[0069] In one or more implementations, the vessel **98** may include a composting bin disposed therein for harvesting the BSFL. The composting bin may include dual walls for retaining heat in the vessel **98**, as illustrated in FIG. **2**. The composting bin and/or the vessel **98** may be in other shapes or sizes. In some implementations, the composting bin may be the vessel **98** as described herein.

[0070] The vessel **98** may have a controlled temperature level and/or humidity level suitable for the BSFL to self-harvest there within. In some implementations, temperature level in the vessel **98** may be controlled to be between approximately 75° F. and 95° F., or any other desired temperature levels. In other implementations, the temperature level in the vessel **98** may be controlled to be between approximately 65° F. and 100° F.

[0071] Referring to FIG. **5**, when the vessel **98** is coupled to the box assembly **82**, an operator may adjust the conditions (e.g., a temperature level, a humidity level, and/or an amount of air) inside the box assembly **82** as well as in the vessel **98** via the electronic control panel **94** so as to provide a suitable condition for the BSFL to self-harvest in the vessel **98**. In some implementations, the temperature level may be set in a range of approximately 65° F. and 100° F., or at any other desired temperature levels. The BSFL may, out of its nature, self-harvest in the vessel **98**. The BSFL may exit the vessel **98** and enter the box assembly **82** through the channel created by the first orifice **100** and the second orifice **102**. When the BSFL are present in the box assembly **82**, they may begin to decompose a body or a portion of a body stored in the box assembly **82**. The BSFL may consume bone marrows of the body or the portion of the body. Depending on the amount of the BSFL available in the box assembly **82**, the conditions within the box assembly **82**, and the size of the body or the portion of the body, the decomposition may last for a few days. After the decomposition is completed, remains of the body or the portion of the body may be removed from the box assembly **82** and returned to family member. The remains may also be donated to science for research studies, planted in the ground (e.g., in a garden), or fed to chicken, fish, birds, or the like. Comparing with the traditional cremation that uses fire, the system **80** described herein for

decomposing bodies offers green cremation, which is a gentle, eco-friendly, and carbon-neutral alternative to the flame-based cremation.

[0072] Although the system **80** as described herein includes one box assembly for green cremation as illustrated herein, more than one such box assemblies may be connected, either side-by-side or end-to-end, when in use. When more than one such box assemblies are connected, each box assembly may use its own electronic control panel and air conditioner to control the conditions there within as well as the conditions in any vessels coupled thereto. Alternatively, the conditions of several such connected box assemblies and any vessels coupled thereto may be controlled by a common electronic control panel and a common air conditioner. Furthermore, when more than one such box assemblies are connected, each box assembly may be coupled to at least one vessel containing BSFL. Alternatively, several such connected box assemblies may share the BSFL coming from one or more common vessel coupled to those box assemblies.

[0073] FIG. **6A** depicts a schematic perspective view of the box assembly of the system as described in FIG. **5**. As shown in FIG. **6A**, the first orifice **100** is provided on the side wall **86** of the box assembly **82**. The first orifice **100** may be configured for the BSFL to enter the box assembly **82** for decomposing a body or a portion of a body. As discussed in FIG. **5**, when more than one vessels are coupled to the side wall **86** of the box assembly **82**, the box assembly **82** may include more than one orifices on the side wall **86** thereof to accommodate the orifices on those vessels. Furthermore, FIG. **6A** shows the cover **96** of the system **80**. The cover **96** may cover or uncover the box opening of the box **84** of the box assembly **82**. In some implementations, the cover **96** may be hingedly connected to the top end **87** of the box **84**. In some other implementations, the cover **96** may be removably disposed onto the top end **87** of the box **84**.

[0074] FIG. **6B** depicts a schematic perspective view of the vessel as described in FIG. **5**. The vessel **98** may include a shape and a size to allow the BSFL to self-harvest there within. For example, the vessel **98** may include a shape and a size to allow the BSFL to establish one or more colonies there within. The vessel **98** may include a slanted wall having a slope angle so as to allow the BSFL to exit the vessel **26**. In some implementations, the slope angle may be between approximately 35° and 45°. When the vessel **98** is coupled to the box assembly **82**, the BSFL may exit the vessel **98** and enter the box assembly **82**. If there is a body or a portion of a body in the box assembly **82**, the BSFL may begin to decompose the body or the portion of the body. In one or more implementations, the vessel **98** may include a composting bin disposed therein for harvesting the BSFL, as illustrated in FIG. **2**. In some implementations, the composting bin may be the vessel **98** as described herein.

[0075] FIG. **6B** shows that the second orifice **102** is provided on the vessel **98**. When the vessel **98** is coupled to the box assembly **82**, the BSFL may exit the vessel **98** via the second orifice **102** and enter the box assembly **82** through the first orifice **100** provided on the at least side wall **86** of the box assembly **82**. As discussed in FIG. **5**, the vessel **98** may include more than one orifices in addition to the second orifice **102**.

[0076] FIG. **6C** depicts a schematic isometric view of the box assembly of the system as described in FIG. **5**. As shown in FIG. **6C**, the insert **88** may be disposed within the box cavity **85**, which separates the box cavity **85** into the first chamber **89** and the second chamber **90**. The insert **88** may be disposed at any location within the box cavity **85**. In some implementations, the insert **88** may be connected to the side wall **86** of the box **84**. In some other implementations, the insert **88** may be connectable to the side wall **86** of the box **84**, in which case the insert **88** may be removable or replaceable. FIG. **6C** further shows that the air conditioner **104** may be disposed in the first chamber **89**. The air conditioner **104** may be configured to control a temperature level and/or humidity level in the box cavity **85** when the air conditioner **104** is in operation. For example, during the decomposition process, the temperature level within the box **84** may rise. The air conditioner **104** may thus be controlled to operate to maintain the temperature level within the desired temperature range as described herein. The second chamber **90** may include a shape and a

size to receive a body or a portion of a body for green cremation. In addition, FIG. 6C shows the first orifice **100** is provided on the side wall **86** of the box assembly **82** for the BSFL entering the box assembly **82** from the vessel **98** when the vessel **98** is coupled to the side wall **86** of the box assembly **82**. Furthermore, FIG. 6C shows the cover **96** of the system **80**. The cover **96** may cover or uncover the box opening of the box **84** of the box assembly **82**. In some implementations, the cover **96** may be hingedly connected to the top end **87** of the box **84**. In some other implementations, the cover **96** may be removably disposed onto the top end **87** of the box **84**.

[0077] FIG. 7 depicts a schematic perspective view of a system for green cremation according to a fourth embodiment of the present disclosure. As shown in FIG. 7, the system **120** may include a box assembly **122** for storing a body or a portion of a body. The body may be a human body, a non-human (e.g., dog or cat) body, or an organ from a human or non-human body. The box assembly **122** may include a box **124** defining a box cavity **125** within the box **124**. The box **124** may further include a bottom end **123** (e.g., a base), a top end **127** defining a box opening connected to the box cavity **125**, and at least one side wall (e.g., side wall **126**) extending from the bottom end **123** to the top end **127**. The side wall **126** at least partially defines the box cavity **125**. The side wall **126** may be made of a thermal insulation material. Although FIG. 7 depicts that the shape of the box assembly **122** is rectangular, it is contemplated that in one or more implementations of the present disclosure, the box assembly **122** may be in any other desired geometric shapes, including, but not limited to, round, square, oval, or polygonal. The box assembly **122** may further include a size to receive a body or a portion of a body therein.

[0078] As shown in FIG. 7, the box assembly **122** may further include an insert **128** disposed within the box cavity **125**, which separates the box cavity **125** into a first chamber **129** and a second chamber **130**. The insert **128** may be disposed at any location within the box cavity **125**. In some implementations, the insert **128** may be connected to the side wall **126** of the box **124**. In some other implementations, the insert **128** may be connectable to the side wall **126** of the box **124**, in which case the insert **128** may be removable or replaceable. As shown in FIG. 7, the insert **128** may include an airway **131** through which air in the box cavity **125** may flow from the first chamber **129** to the second chamber **130**, or vice versa. The airway **131** may be an opening disposed in the insert **128**. In some implementations, a vent may be used to cover the opening and allow for the air in the box cavity **125** to flow between the first chamber **129** and the second chamber **130**. The box assembly **122** may further include an air conditioner (similar to **104** in FIG. 6C) disposed in the first chamber **129**. The air conditioner may be configured to control a temperature level and/or a humidity level in the box cavity **125** when it is in operation. In some implementations, the air conditioner may include one or more devices that control other environmental conditions within the box cavity **125** (e.g., moisture control and odors). The second chamber **130** may include a shape and a size to receive a body or a portion of a body for green cremation.

[0079] Referring to FIG. 7, the box assembly **122** may further include an air vent **132** disposed on the side wall **126** of the box **124** thereof. The air vent **132** may allow air to be exchanged between the box cavity **125** and an ambient environment (i.e. outside the system **120**). For example, the air vent **132** may be configured for the air inside the box cavity **125** to exit the box assembly **122** or may be configured for the air in the ambient environment to enter the box assembly **122**.

[0080] In some implementations, the box assembly **122** may further include an ozone generator disposed on the side wall **126** of the box **124**. The ozone generator (not shown) may be configured to remove odors from the box cavity **125**. For example, the ozone generator may be disposed in an exhaust chamber with a one-way air valve. In such an implementation, the ozone generator is configured to remove odors from the box assembly **122** without exposing the insects in the box assembly **122** to ozone.

[0081] The box assembly **122** may also include an electronic control panel **134** connected thereto and configured to control the operation of the system **120**, including the operation of the air

conditioner. In some implementations, the electronic control panel **134** may be connected to the side wall **126** of the box **124**. According to one or more implementations of the present disclosure, the electronic control panel **134** may be a display interface, through which an operator may input operating instructions to the system **120** for the purpose of controlling the conditions (e.g., a temperature level, a humidity level, or an air amount) within the box assembly **122**. Upon a request by the operator, the electronic control panel **134** may display images or videos showing a scene of the box cavity **125**. This allows the operator to monitor the operation of the system **120** in real time or near real time. The electronic control panel **134** may further electronically communicate with a remote server, such as a remote storage cloud, a remote control center, or a mobile phone of the operator or of any person of interest (e.g., a family member of the body or the portion of the body). The electronic control panel **134** may also, by wires or wirelessly, connect to at least one another electronic control panel of another system of this kind. The connection between or among such systems may provide additional features, including, but not limited to, data sharing between or among the systems, remote controls of the systems, or alert notifications among the systems. For example, if there is a technical problem with one of the connected systems, the system having the technical problem may generate an alert, which may be sent to other connected systems or to a technician or an operator who may then fix the technical problem.

[0082] The electronic control panel **134** may also leverage the Internet of Things (IoT), Bluetooth, or Wi-Fi to remotely connect to, enable, and control one or more functionalities of the system **120**. For example, the electronic control panel **134** may be configured to present a display interface on a remote device, through which a user may input operating instructions to the system **120** for the purpose of controlling the conditions within the box assembly **122** (e.g., a temperature, a humidity level, or an air amount).

[0083] Referring to FIG. 7, the system **120** may further include a cover **136** which may cover or uncover the box opening of the box **124** of the box assembly **122** thereof. In some implementations, the cover **136** may be hingedly connected to the top end **127** of the box **124**. In some other implementations, the cover **136** may be removably disposed onto the top end **127** of the box **124**. The system **120** may be in an open state when the cover **136** uncovers the box cavity **125**. When the system **120** is in the open state, the operator may place a body or a portion of the body into the box assembly **122** (e.g., the second chamber). The system **120** may be in a closed state when the cover **136** covers the box cavity **125**. When the system **120** is in the closed state, the operator may input operating instructions to the system **120** to control the conditions (e.g., a temperature level, a humidity level, or an air amount) within the box assembly **122** for green cremation.

[0084] To carry out green cremation according to this embodiment of the present disclosure, the system **120** may further include a vessel **138** coupled to the side wall **126** of the box assembly **122**. In some implementations, the vessel **138** may be glued, screwed, or hooked to the side wall **16** of the box assembly, or any other desired coupling mechanisms.

[0085] As shown in FIG. 7, the vessel **138** may be coupled to an inner surface of the side wall **126** of the box assembly **122**. In some implementations, the vessel **138** may be integrated with the box assembly **122**. In some other implementations, the vessel **138** may be removably attached to the box assembly **122**. It is contemplated that although FIG. 7 shows one vessel coupled to the side wall **126** of the box assembly **122**, more than one vessels may be similarly coupled thereto. It is further contemplated that although FIG. 7 shows that the vessel **138** is coupled into the second chamber **130**, the vessel **138** may also be coupled into the first chamber **129**. The vessel **138** may include an orifice **140**, which may be configured for BSFL to exit the vessel **138** to decompose the body or the portion of the body stored in the box assembly **122**. In some implementations, the vessel **138** may include more than one orifices in addition to the orifice **140** shown in FIG. 7.

[0086] The vessel **138** may contain BSFL. The vessel **138** may include a shape and a size to allow the BSFL to self-harvest there within. For example, the vessel **138** may include a shape and a size

to allow the BSFL to establish one or more colonies there within. The vessel **138** may include a slanted wall having a slope angle so as to allow the BSFL to exit the vessel **138**. In some implementations, the slope angle may be between approximately 35° and 45°. Referring to FIG. 7, when the vessel **138** is coupled to the box assembly **122**, the BSFL may exit the vessel **138** and enter the box assembly **122**. If there is a body or a portion of a body in the box assembly **122**, the BSFL may begin to decompose the body or the portion of the body. To escape the heat produced by the decomposing body, the BSFL may be motivated to exit the second chamber **130** of the box cavity **125**. In some implementations, the second chamber **130** has slanted walls with a slope angle so as to allow the BSFL to escape the second chamber **130** and then be collected in the bottom of the first chamber **129** of the box cavity **125**.

[0087] In some implementations, the vessel **138** may include multiple structures. For example, the BSFL may self-harvest from a first structure of the vessel **138** into a second vessel of the vessel **138** using an escape ramp having an angle of no more than 60°. The BSFL may then exit the second structure and enter the box assembly **122**.

[0088] In one or more implementations, the vessel **138** may include a composting bin disposed therein for harvesting the BSFL. The composting bin may include dual walls for retaining heat in the vessel, as illustrated in FIG. 2. The composting bin and/or the vessel **138** may be in other shapes or sizes if necessary. In some implementations, the composting bin may be the vessel **66** as described herein.

[0089] The vessel **138** may have a controlled temperature level and/or humidity level suitable for the BSFL to self-harvest there within. In some implementations, the temperature level in the vessel **138** may be controlled to be between approximately 75° F. and 95° F., or any other desired temperature levels. In other implementations, the temperature level in the vessel **66** may be controlled to be between approximately 65° F. and 100° F.

[0090] Referring to FIG. 7, when the vessel **138** is coupled to the box assembly **122**, an operator may adjust the conditions (e.g., a temperature level, a humidity level, and/or an amount of air) inside the box assembly **122** as well as in the vessel **138** via the electronic control panel **134** so as to provide a suitable condition for the BSFL to self-harvest in the vessel **138**. In some implementations, the temperature level may be set in a range of approximately 65° F. and 100° F., or at any other desired temperature levels. The BSFL may, out of its nature, self-harvest in the vessel **138**. When the BSFL become mature, they may exit the vessel **138** and enter the box assembly **122** through the orifice **140**. When the BSFL are present in the box assembly **122**, they may begin to decompose the body or the portion of the body stored in the box assembly **122**. The BSFL may consume bone marrows of the body or the portion of the body. Depending on the amount of the BSFL available in the box assembly **122**, the conditions within the box assembly **122**, and the size of the body or the portion of the body, the decomposition may last for a few days. After the decomposition is completed, remains of the body or the portion of the body may be removed and returned to family members. The remains may also be donated to science for research studies, planted in the ground (e.g., in a garden), or fed to chicken, fish, birds, or the like. Comparing with the traditional cremation that uses fire, the system **120** described herein for decomposing bodies offers green cremation, which is a gentle, eco-friendly, and carbon-neutral alternative to the flame-based cremation.

[0091] Although the system **120** as described herein includes one box assembly for green cremation as illustrated herein, more than one such box assemblies may be connected, either side-by-side or end-to-end, when in use. When more than one such box assemblies are connected, each box assembly may use its own electronic control panel and air conditioner to control the conditions there within as well as the conditions in any vessels coupled thereto. Alternatively, the conditions of several such connected box assemblies and any vessels coupled thereto may be controlled by a common electronic control panel and a common air conditioner. Furthermore, when more than one such box assemblies are connected, each box assembly may be coupled to at least one vessel

containing BSFL. Alternatively, several such connected box assemblies may share the BSFL coming from one or more common vessel coupled to those box assemblies.

[0092] No part of the description in this application should be read as implying that any particular element, step, or function is an essential element that must be included in the claim scope. The scope of patented subject matter is defined only by the claims. Moreover, none of the claims is intended to invoke 25 U.S.C. § 104 (f) unless the exact words “means for” are followed by a participle.

[0093] The foregoing description, for purposes of explanation, use specific nomenclature to provide a thorough understanding of the described embodiments. However, it should be apparent to one skilled in the art that the specific details are not required to practice the described embodiments. Thus, the foregoing descriptions of specific embodiments are presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the described embodiments to the precise forms disclosed. It should be apparent to one of ordinary skill in the art that many modifications and variations are possible in view of the above teachings.

[0094] The above discussion is meant to be illustrative of the principles and various embodiments of the present disclosure. Once the above disclosure is fully appreciated, numerous variations and modifications will become apparent to those skilled in the art. It is intended that the following claims be interpreted to embrace all such variations and modifications.

Claims

1. A system for cremating a body, the system being situated in an environment whose conditions are controllable, the system comprising: a box assembly for storing the body or a portion of the body, the box assembly including: a box defining a box cavity within the box, a bottom end, a top end defining a box opening connected to the box cavity, and at least one side wall extending from the bottom end to the top end, and an air conditioner disposed in the box cavity and configured to control a temperature level or a humidity level in the box cavity when the air conditioner is in operation; and a vessel coupled to the at least one side wall of the box of the box assembly, wherein the vessel includes insect larvae that self-harvest, exit the vessel, and enter the box assembly to decompose the body or the portion of the body stored in the box assembly.
2. The system of claim 1, wherein the at least one side wall of the box includes a first orifice that is configured for the insect larvae to enter the box assembly.
3. The system of claim 2, wherein the vessel is coupled to an outer surface of the at least one side wall of the box, and the vessel includes a second orifice that is configured for the insect larvae to exit the vessel.
4. The system of claim 1, wherein the vessel is coupled to an inner surface of the at least one side wall of the box, and the vessel includes an orifice that is configured for the insect larvae to exit the vessel and enter the box assembly.
5. The system of claim 1, wherein the at least one side wall is made of a thermal insulation material.
6. The system of claim 1, wherein the temperature level is controlled to be between approximately 65° F. and 100° F.
7. The system of claim 1, wherein the box assembly further includes an insert disposed within the box cavity, the insert separating the box cavity into a first chamber and a second chamber, wherein the air conditioner is disposed in the first chamber, and the body or the portion of the body is stored in the second chamber.
8. The system of claim 7, wherein the insert further includes an airway through which air in the box cavity flows between the first and second chambers.
9. The system of claim 1, wherein the box assembly further includes: an isolated air chamber disposed on the at least one side wall of the box, and an ozone generator disposed in the isolated air

chamber and configured to control odors escaping from the box cavity to the environment in which the system is situated.

10. The system of claim 1, further comprising an electronic control panel configured to control the operation of the air conditioner.

11. A system for cremating a body, the system comprising: a box assembly for storing the body or a portion of the body, the box assembly including: a box defining a box cavity within the box, a bottom end, a top end defining a box opening connected to the box cavity, and at least one side wall extending from the bottom end to the top end, and an air conditioner disposed within the box cavity and configured to control a temperature level or a humidity level in the box cavity when the air conditioner is in operation; a vessel coupled to the at least one side wall of the box of the box assembly, wherein the vessel includes insect larvae that self-harvest, exit the vessel, and enter the box assembly to decompose the body or the portion of the body stored in the box assembly; and a cover hingedly connected to or removably disposed onto the top end of the box of the box assembly, wherein the system is in an open state when the cover uncovers the box cavity of the box, and in a closed state when the cover covers the box cavity of the box.

12. The system of claim 11, wherein the at least one side wall of the box includes a first orifice that is configured for the insect larvae to enter the box assembly.

13. The system of claim 12, wherein the vessel is coupled to an outer surface of the at least one side wall of the box, and the vessel includes a second orifice that is configured for the insect larvae to exit the vessel.

14. The system of claim 11, wherein the vessel is coupled to an inner surface of the at least one side wall of the box, and the vessel includes an orifice that is configured for the insect larvae to exit the vessel and enter the box assembly.

15. The system of claim 11, wherein the at least one side wall is made of a thermal insulation material.

16. The system of claim 11, wherein the temperature level is controlled to be between approximately 65° F. and 100° F.

17. The system of claim 11, wherein the box assembly further includes an insert disposed within the box cavity, the insert separating the box cavity into a first chamber and a second chamber, wherein the air conditioner is disposed in the first chamber, and the body or the portion of the body is stored in the second chamber.

18. The system of claim 17, wherein the insert further includes an airway through which air in the box cavity flows between the first and second chambers.

19. The system of claim 11, wherein the box assembly further includes: an isolated air chamber disposed on the at least one side wall of the box, and an ozone generator disposed in the isolated air chamber and configured to control odors escaping from the box cavity to an ambient environment that is outside the system.

20. The system of claim 11, further comprising an electronic control panel configured to control the operation of the air conditioner.
